

An electrochromic device based on all-in-one polymer gel through in-situ thermal polymerization[☆]

Sheng-Yuan Kao^a, Chung-Wei Kung^a, Hsin-Wei Chen^a, Chih-Wei Hu^a, Kuo-Chuan Ho^{a,b,*}

^a Department of Chemical Engineering, National Taiwan University, Taipei 10617, Taiwan

^b Institute of Polymer Science and Engineering, National Taiwan University, Taipei 10617, Taiwan

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ABSTRACT

A self-standing all-in-one electrochromic gel utilizing phenyl viologen (PV) and N,N,N',N'-tetramethyl-1,4-phenylenediamine (TMPD) as the electrochromic materials is prepared by incorporating a thermocured cross-linked polymer matrix. The thermal-cured polymer matrix can prevent the agglomeration of PV happening on the electrode surface, resulting in the excellent write-erase ability of the obtained gelled ECD. Moreover, the cross-linker which possesses the long propoxylated arms leads to the easier diffusion for both PV and TMPD in the obtained polymer network. As a result, the obtained ECD exhibits a large transmittance change ($\Delta T > 67\%$ at 620 nm) within a short response time (< 3.5 s) under an extremely low driving voltage (0.5 V). Besides, this device remains over 55% of ΔT after 10,000 cycles of switching.

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1. Introduction

Electrochromic devices (ECDs) change their color through the electrochemical manipulation of redox states of electrochromic (EC) materials [1–3]. Such devices have already been proposed in several potential applications such as electrochromic smart windows [4,5] anti-glare rearview mirrors [6], electrochromic helmet visors [7] and displays [8]. Since the smart window and display devices using suspended particles or liquid crystals often require a potential bias higher than 5 V to exhibit sufficient change in color, ECDs have attracted much attention due to their relatively low driving voltage (1–4 V) [4]. In addition to their low driving voltage, ECDs could be integrated with photovoltaic to fabricate the self-powered energy-saving electronic devices [9,10].

Among all electrochromic materials, viologens (1,1'-disubstituted-4,4'-bipyridiniums) have been considered as attractive candidates since they exhibit significant change in transmittance [11–13]. Besides, some viologen-based ECDs even possess driving voltage within 1 V [14], which makes this class of electrochromic materials even more attractive in terms of energy-saving.

Nevertheless, most viologen-based ECDs possess poor long-term stability, which is attributed to the undesirable side reactions such as dimerization [15,16] and comproportionation [17] occurring during operation. Furthermore, it has been reported that the colored state of viologens has a lower solubility compared to the bleached state. As a result, high concentration of colored viologen tends to form agglomerates on the electrode surface [18]; such deposited agglomerates would begin to crystalize if the coloring process lasts longer [19]. The generated agglomerates hinder the bleaching reaction of viologens, leading to the poor write-erase ability and poor cycling stability of the obtained ECDs [20]. Moreover, most ECDs utilizing viologens are mostly solution-type [14]; such solution-type ECDs usually suffer from leakage of electrolyte, which also harms the stability and brings safety concerns [2].

The electrochromic process of 1,1'-diphenyl-4,4'-bipyridinium (phenyl viologen, PV) can be operated at an extremely low potential bias of about 0.5 V, with the help of N,N,N',N'-tetramethyl-p-phenylenediamine (TMPD) as the redox couple on the counter electrode. The TMPD/TMPD⁺ redox couple can provide/extract electrons to and from PV²⁺/PV^{•+} respectively during the operation of the device, therefore making the coloring/bleaching process easier under this narrow potential bias [14]. Although PV has been discovered and synthesized for several decades [21,22], there is not any reported study applied such material for practical ECDs due to the nature of its poor cycling stability mentioned

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* Corresponding author at: Department of Chemical Engineering, National Taiwan University, Taipei 10617, Taiwan. Tel.: +886 2 2366 0739; fax: +886 2 2362 3040.

E-mail address: kcho@ntu.edu.tw (K.-C. Ho).

above. Compared to other viologens such as coil-moiety bearing heptyl viologens, which have been widely utilized in ECDs [23], the rod-like linear molecules including PV are expected to form agglomerate on the electrode surface much easier [24–26]; this disadvantage thus limits the use of PV for ECDs.

Several strategies have been proposed to prevent the agglomeration of viologens on the electrode surface in ECDs. To effectively inhibit the intermolecular interaction between viologens, Monk et al. added electron mediators like ferrocyanide/ferricyanide [23,27] while Yasuda et al. included cyclodextrin with viologen to generate a complex that prevents deposition of viologen [28]. Wrighton et al. proposed a viologen-based polymer which was covalently anchored on the electrode surface [29], and Walder et al. anchored the viologen molecules onto a porous modified electrode [30]. All these strategies have successfully eliminated the agglomeration of viologens. However, the ECDs proposed in the above-mentioned literatures still used liquid electrolyte, which might cause the undesired leakage problem. For the first time, Chidichimo et al. [31] introduced substantial amount of poly(vinyl formal) into the viologen-based electrochromic system to obtain a thermal plastic laminable electrochromic gel. This electrochromic gel has successfully prevented the leakage problem. However, such ECDs possess much higher potential bias (> 1.5 V) and longer response time to reveal significant color contrast. To date, very few viologen-based ECDs acquire all the merits such as low-driving voltage, good stability, and high transmittance change simultaneously.

Herein, the electrochromic polymer gel was incorporated into the viologen-based electrochromic system by utilizing a simple in-situ thermal curing method. PV and TMPD were served as the cathodic and anodic electrochromic materials, respectively. Methyl methacrylate (MMA) was used as a gelling agent while azobisisobutyronitrile (AIBN) was employed as a thermal curing initiator as suggested in literature [32]. Propoxylated trimethylolpropane triacrylate (PTPTA) was utilized as the cross-linker since it can provide a more robust network in the electrochromic gel than the two-acrylate cross-linker [33]. The PV and TMPD systems give the lowest driving voltage (0.5 V) for leakage-free, all-in-one, EC gel to date while exhibiting excellent cycling stability. The resulted thermal-cured polymer gel offers remarkably high transparency and superior ionic conductivity, making it an ideal polymer electrolyte for ECDs. A large transmittance change ($> 65\%$) was achieved; even the ECD was incorporated with the polymer network.

2. Experimental

2.1. Chemicals

Phenyl viologen dichloride ($PVCl_2$, $> 97\%$) was purchased from Tokyo Chemical Industry (TCI). Sodium tetrafluoroborate ($NaBF_4$, 98%), TMPD (99%), MMA (99%), propoxylated trimethylolpropane triacrylate (PTPTA), and tetra-*n*-butylammonium tetrafluoroborate ($TBABF_4$, 99%) were purchased from Sigma-Aldrich. Propylene carbonate (PC, 99%) was purchased from Alfa Aesar while AIBN (99.7%) was purchased from UniRegion. Deionized water (DIW) with the resistance of $18.2\text{ M}\Omega$ was utilized throughout the experiment. Except PC which was dehydrated by $4\text{ \AA}$ molecular sieve (Acros) for at least overnight, all other chemicals were used as received without further purification. Indium tin oxide coated conducting glass (ITO, Solaronix SA, $R_{sh}=7\text{ }\Omega/\square$) was used as the conducting substrate throughout this work. Since $PVCl_2$ shows poor solubility in PC, the chloride anions were substituted to tetrafluoroborate (BF_4^-) anions by the approach reported in literatures [14,34]. Briefly, 0.3 g $PVCl_2$ was dissolved in minimal amount of $90\text{ }^\circ\text{C}$ DIW. This solution was immediately mixed with 3.75 ml of 9.4 mM $NaBF_4$ aqueous solution under vigorously stirring. The

resulted yellowish $PV(BF_4)_2$ precipitant was then washed thoroughly with ethanol/DIW solution. After drying, the obtained $PV(BF_4)_2$ powder exhibits nice solubility in organic solvents such as PC.

2.2. Preparation of the electrochromic gel

To obtain the all-in-one electrochromic gel, MMA was utilized as the monomer for polymer electrolyte and PTPTA was employed as a cross-linker. 5% (v/v) of MMA and 5% (v/v) of PTPTA were mixed homogeneously with equimolar electrochromic materials (0.05 M $PV(BF_4)_2$ and 0.05 M TMPD) and 0.5 M $TBABF_4$ along with the plasticizer PC to form an all-in-one electrochromic gel precursor. After blending, trace amount (8.2 mg) of the thermal cross-linking initiator (AIBN) was dissolved into the precursor. This precursor was then stored in vials or injected in sealed devices (illustrated in Section 2.3). Thereafter, the electrochromic gel precursor was heated at $90\text{ }^\circ\text{C}$ for ca. 40 min for thermal curing.

2.3. Fabrication of ECDs

In this study, two types of ECDs are fabricated, namely, the gelled ECD (g-ECD) and the conventional ECD (c-ECD). The g-ECD was assembled by two ITO substrates with the equal size of $3.0 \times 4.0\text{ cm}^2$ that acted as the working and counter electrodes. Prior to use, the ITO substrates were cleaned by isopropyl alcohol and then subjected to ozone treatment for 20 min. One of the ITO glasses was drilled with two holes for precursor injection. The cell gap was controlled by one layer of DuPont $60\text{ }\mu\text{m}$ Surlyn[®] (Solaronix S.A., Aubonne, Switzerland). The Surlyn[®] ionomer resin was first cut into $2.0 \times 2.0\text{ cm}^2$ frame and cleaned by ethanol before laminated between the two ITO substrates. The laminated Surlyn[®] resin was then heated and pressed for sealing the device. After sealing the device, the obtained precursor mentioned in Section 2.2 was injected into the ECD through the drilled holes. The g-ECD was then obtained through subjected to $90\text{ }^\circ\text{C}$ heating for 40 min to cure the electrolyte. For comparison, the c-ECD without using polymer electrolyte was also fabricated by injecting the electrolyte containing 0.05 M $PV(BF_4)_2$, 0.05 M TMPD and 0.5 M $TBABF_4$. Finally, the holes on all ECDs were further sealed by epoxy.

2.4. Measurements

Electrochemical data was obtained by using a potentiostat/galvanostat (Autolab, model PGSTAT30). Spectro-electrochemical data was acquired by in-situ electrochemical and optical measurements, which were conducted on a potentiostat/galvanostat in conjunction with a spectrophotometer (Ocean Optics, DH-2000-BAL). Electrochemical quartz crystal microbalance (EQCM) was done by utilizing a potentiostat/galvanostat connected with a quartz crystal microbalance oscillator (Seiko EG&G, QCA917). In this setup, 8.88-MHz AT-cut quartz crystal (QA-AM9-PT) with a Pt electrode (with an active surface area of 0.196 cm^2) served as the working electrode while homemade Ag/Ag^+ and Pt foil ($1 \times 4\text{ cm}^2$) served as the reference and counter electrodes, respectively.

3. Results and discussion

3.1. Properties of the all-in-one EC gel

The cyclic voltammetric (CV) measurements of $PV^{2+}/PV^{\bullet+}$ and $TMPD^+/TMPD^0$ are presented in Fig. 1(a) and (b) respectively. When applying the negative potential sweep in Fig. 1(a), green color appears near the working electrode, indicating the formation

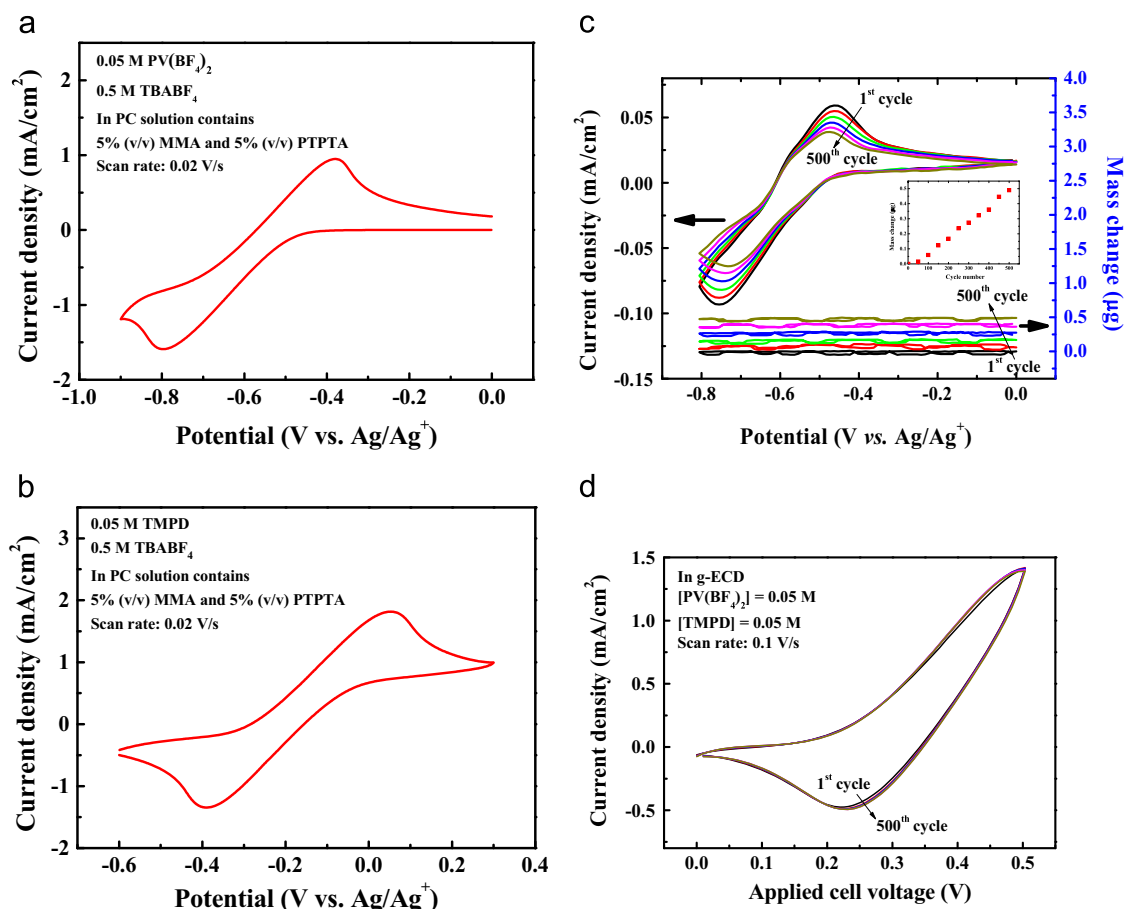
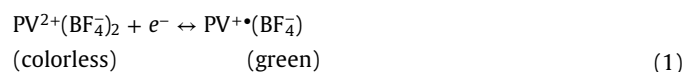


Fig. 1. Cyclic voltammetric measurement of (a) PV and (b) TMPD in a three-electrode system. (c) Current density and accumulated mass change of the Pt electrode recorded during 500 cycles of CV scan in the solution containing 1 mM PV(BF₄)₂ and 10 mM TBABF₄. Inset shows the variation of accumulated mass obtained in (c) in response to cycling number. (d) 500-cycle CV curves of the g-ECD scanned between 0 and 0.5 V at a scan rate of 0.1 V/s. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

of PV^{•+} as illustrated by Eq. (1).



This green PV^{•+} species would gradually diffuse back to the bulk solution soon after its formation. Therefore, a relatively small oxidation peak of PV^{•+} is observed. On the other hand, violet TMPD^{•+} was generated during the positive potential sweep in Fig. 1(b) as shown in Eq. (2).



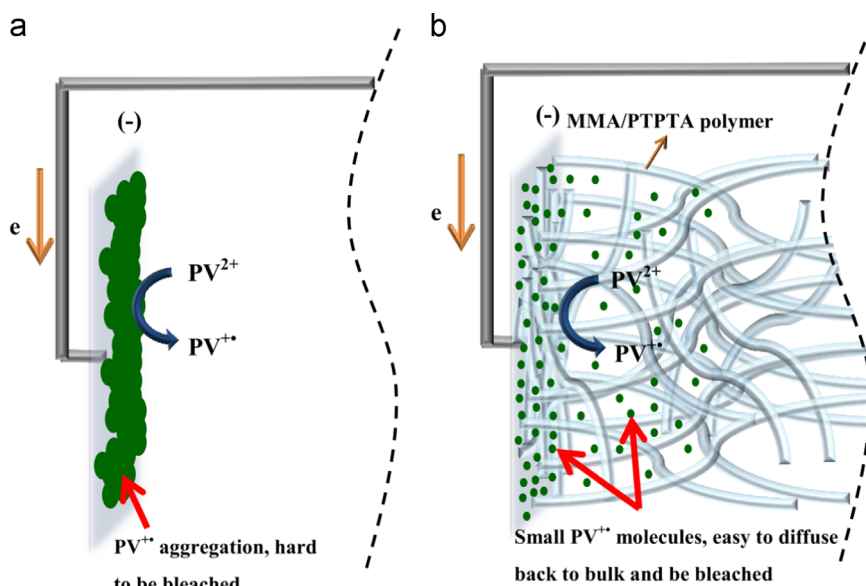
The diffusion of TMPD^{•+} back in the bulk solution can also be observed, which explains a weaker reduction peak compared to the oxidation peak in Fig. 1(b).

Before utilizing the EC polymer gel, the electrochemical property of PV was further investigated by 500-cycle cyclic voltammetric (CV) measurement coupled with EQCM; the experiment was done in a liquid solution containing 1.0 mM PV(BF₄)₂ and 10.0 mM TBABF₄ in PC (Fig. 1(c)). The Pt electrode was scanned between 0 and -0.8 V (vs. Ag/Ag⁺), which is sufficiently wide for the bleaching and coloring reactions of PV^{•+}(BF₄) and PV²⁺(BF₄)₂, respectively. However, significant decrease in the peak current density can be observed in the CV curves shown in Fig. 1(c), which infers the occurrence of irreversible reaction caused by the

agglomeration of PV^{•+}(BF₄) during cycling. The agglomeration of PV happening during its reduction was verified by EQCM analysis measured simultaneously during the CV scan. As shown in Fig. 1(c) and its inset, the mass accumulation on the Pt electrode continuously increased during the CV scan, which further confirms the aggregation of PV^{•+}(BF₄) on the electrode surface. The aggregation of PV^{•+}(BF₄) hinders the oxidation of PV^{•+}(BF₄), which leads to the poor long-term stability in both redox current and transmittance change. Therefore, preventing PV^{•+}(BF₄) from aggregation is necessary in order to incorporate PV in a stable ECD.

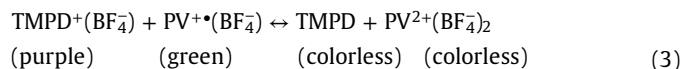
The thermally cured PV/TMPD-based all-in-one polymer gel was thus developed in order to prevent the aggregation of PV^{•+}(BF₄). The 500-cycle CV curve of the all-in-one electrochromic gel obtained under a two-electrode system (g-ECD) is shown in Fig. 1(d). It can be seen that the CV curve of the polymer gel keeps stable during 500 cycles of scan, indicating that the agglomeration of PV did not happen in the polymer gel during the scan. The thermally cured gel electrolyte provides a polymer framework strongly adhering on the electrode surface, which effectively suppress the aggregation of PV^{•+}(BF₄) (Scheme 1(a) and (b)).

In Fig. 1(d), the current density rises when the potential bias reaches over 0.3 V, indicating that the reduction of PV²⁺ on the counter electrode and the oxidation of TMPD on the working electrode occur simultaneously. Meanwhile, these colored species formed at both electrodes would diffuse back to the bulk solution driven by the concentration gradient. The recombination reaction occurs when these two colored species encounter, regenerating



Scheme 1. Mechanism for preventing the agglomeration of PV molecules by utilizing the all-in-one EC polymer gel. (a) In c-ECD (liquid solution), (b) In g-ECD (gel electrolyte).

the colorless PV^{2+} and TMPD. The recombination reaction can be written as follows:



By combining Eqs. (1), (2), and (3), the working principle of the g-ECD can be constructed (Scheme S1). In other words, when the potential bias of the g-ECD was scanned from 0.5 V back to 0 V, the coloring reactions (Eqs. (1) and (2)) stopped while the recombination process in the bulk solution (Eq. (3)) continued.

Such self-bleaching characteristic can be demonstrated in both c-ECD and g-ECD. To do so, these two as-prepared ECDs were initially subjected to a potential bias of 0.5 V for 15 s and then left under the open-circuit condition. Their transmittance data were collected at 580 nm as presented in Fig. S1. It took ca. 9.5 s for the colored g-ECD to gradually return to its original bleaching state. On the other hand, it takes ca. 7.1 s for the transmittance of the c-ECD to reach its saturated bleaching state. A slightly longer time is required for the g-ECD to reach its transmittance at the saturated bleaching state suggesting a slightly lower diffusion process of the colored species compared to that of the c-ECD. It is worthwhile to mention that the transmittance of the c-ECD cannot return to its original bleaching state due to the formation of the colored agglomeration (as suggested in Fig. 1(c)). In contrast, the effective self-erasing characteristics of the g-ECD results in the stable dynamic feature, as shown in Fig. S1 and the CV curve in Fig. 1(d).

The UV–visible absorbance spectra of the g-ECD measured at various potential biases are presented in Fig. 2. Under the short-circuit condition (0 V), the absorbance is lower than 0.2 throughout the whole visible region, suggesting that high transparency can be achieved even with the existence of the polymer network. As the potential bias reaches beyond 0.2 V, the absorbance of the g-ECD increases significantly in the whole visible region. This observation is in accordance with the rise in the current density shown in Fig. 1(d), inferring that the reduction of PV^{2+} and oxidation of TMPD take place simultaneously. As the potential bias reaches 0.5 V, the absorbance becomes saturated, with four peaks located at 430, 580, 620, and 715 nm. According to the observation reported in literature [21], peaks at 430 and 715 nm can be

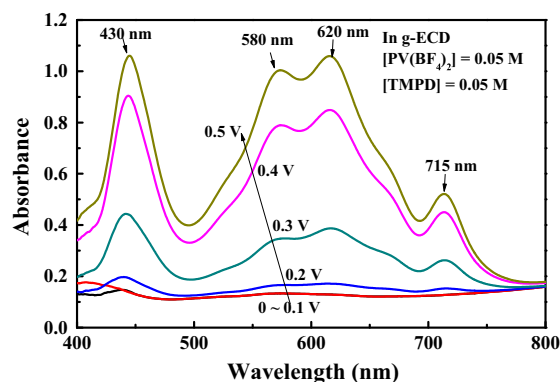


Fig. 2. UV–visible absorbance spectra of the g-ECD measured at various potential biases ranging from 0 to 0.5 V.

assigned to $PV^{+}(\text{BF}_4^-)$. Notably, $PV^{+}(\text{BF}_4^-)$ also exhibits another broad peak spreading from 500 to 700 nm. On the other hand, TMPD^+ possesses two absorbance peaks at 580 and 620 nm as suggested in literature [14]. Therefore, the absorbance peaks located at 580 and 620 nm observed in Fig. 2 are contributed both from $PV^{+}(\text{BF}_4^-)$ and TMPD^+ .

The photographs of the self-standing thermal-cured EC gel are presented in Fig. 3(a). It can be seen that the bulk polymer gel exhibits a dark brown color (inset of Fig. 3(a)). However, when the EC gel is thermally cured inside an ECD configuration, very high transparency can be achieved since the thickness of the EC gel is limited to 60 μm (Fig. 3(b)). But even if it is a thin electrochromic gel, upon the potential bias of 0.5 V, obvious dark green color of the EC gel can be inspected (Fig. 3(c)). In the next section, quantitative characterizations of the transmittance response are done at both 580 and 620 nm, at which the g-ECD exhibits the largest transmittance change (ΔT); appreciably, these two wavelengths are also located within the region that is most sensitive for human eyes.

3.2. EC performance of the g-ECD and c-ECD

Fig. 4(a) and (b) shows the dynamic transmittance curves of the g-ECD measured during the potential switching at 580 and

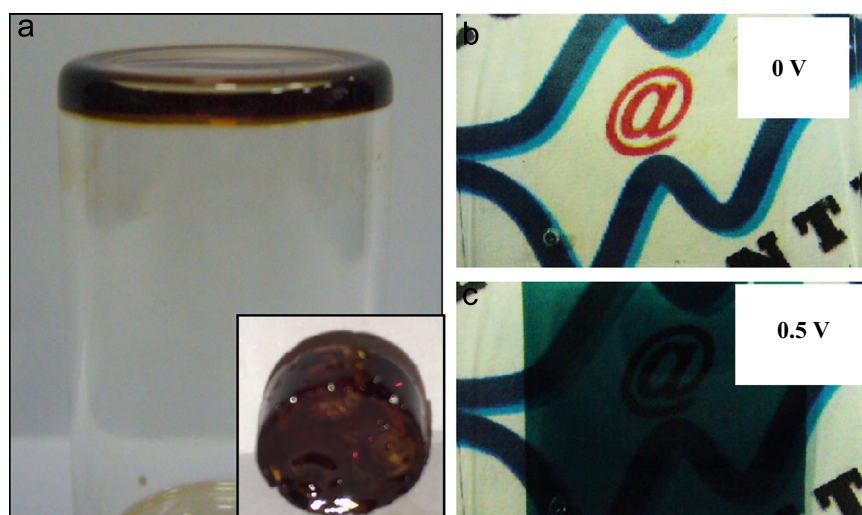


Fig. 3. Photographs of (a) the thermal-cured self-standing electrochromic gel, (b) thermal-cured g-ECD at bleached state, and (c) thermal-cured g-ECD at colored state. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

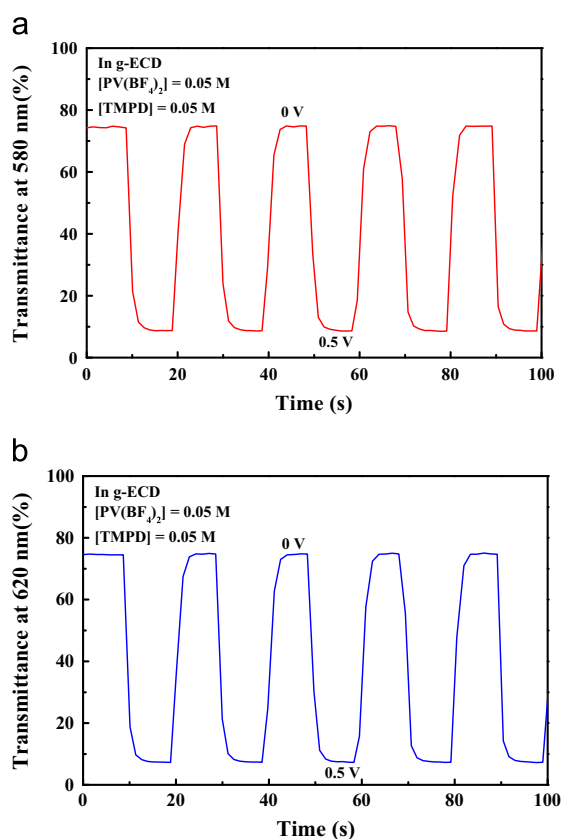


Fig. 4. Dynamic transmittance curves of the g-ECD recorded at (a) 580 and (b) 620 nm. The potential bias was switched between 0 and 0.5 V with the interval of 10 s.

620 nm, respectively. The interval of each potential step was set to be 10 s, which is sufficient for the g-ECD to be fully colored and bleached. It can be seen that the values of ΔT for the g-ECD at both wavelengths are stable during the potential switching. The quantitative performance of the g-ECD is summarized in Table 1. The bleached and colored states of the g-ECD exhibit a transmittance (T_b and T_c) of 74.4% and 7.3% at 620 nm, respectively. Namely, a large ΔT of 67.1% can be achieved at an extremely low driving voltage (0.5 V). High ΔT value of 65.6% is also observed at another absorbance peak located at 580 nm. The bleaching and coloring

Table 1

Quantitative EC performances of the gelled ECD (g-ECD) at 580 and 620 nm.

Potential window (V)	λ (nm)	T_b (%)	T_c (%)	τ_b (s)	τ_c (s)	ΔT (%)
0–0.5	580	74.3	8.7	3.2	2.8	65.6
0–0.5	620	74.4	7.3	3.4	2.7	67.1

times (τ_b and τ_c) are defined as the time required to reach 95% of the saturated ΔT . All these response times are shorter than 3.5 s (see Table 1). For comparison, the dynamic transmittance curves of c-ECD measured at 580 and 620 nm are shown in Fig. S2. It can be seen that the c-ECD shows slightly shorter response times and even higher ΔT values (over 70%) compared to the g-ECD at the first cycle of switching; however, the values of its ΔT at both wavelengths decrease continuously after the first cycle of switching. The quantitative performance of the c-ECD at the first cycle of switching is summarized in Table S1. The g-ECD and c-ECD show similar response time, which infers that little hindrance is subjected to the coloring and bleaching processes by the polymer network in the g-ECD. The long propoxylated arms of PTPTA provide large pore sizes in the polymer framework, which alleviates the hindrance in migration and diffusion of the electrochromic materials in the g-ECD.

The write–erase abilities of both ECDs are presented in Fig. 5. The g-ECD reveals excellent write–erase ability; its ΔT remains 94.2% and 93.3% of the original value after 1000 cycles of switching (20,000 s in total) at 580 and 620 nm, respectively (Fig. 5(a) and (b)). After 3000 cycles of switching, the g-ECD still maintains 92.8% and 92.7% of its initial ΔT values at 580 and 620 nm, respectively (60,000 s in total, Fig. S3(a) and (b)). Namely, the values of ΔT at both wavelengths remain over 60% even after 3000 cycles of potential switching. In contrast, the ΔT values of the c-ECD drastically decrease in the beginning and remain only 32.0% and 24.5% of the original values after 1000 cycles of switching at 580 and 620 nm, respectively (20,000 s in total, Fig. 5(c) and (d)). Even though the c-ECD shows slightly higher ΔT values than those of the g-ECD in the beginning, the poor cycling stability of c-ECD restricts the possibility for its application in smart window or display applications.

The low driving voltage of the g-ECD offered its advantage from the energy-saving point of view, although to this stage, only $2 \times 2 \text{ cm}^2$ of an ECD was fabricated. Scaling up the g-ECD is possible since the in-situ thermal curing reaction would occur

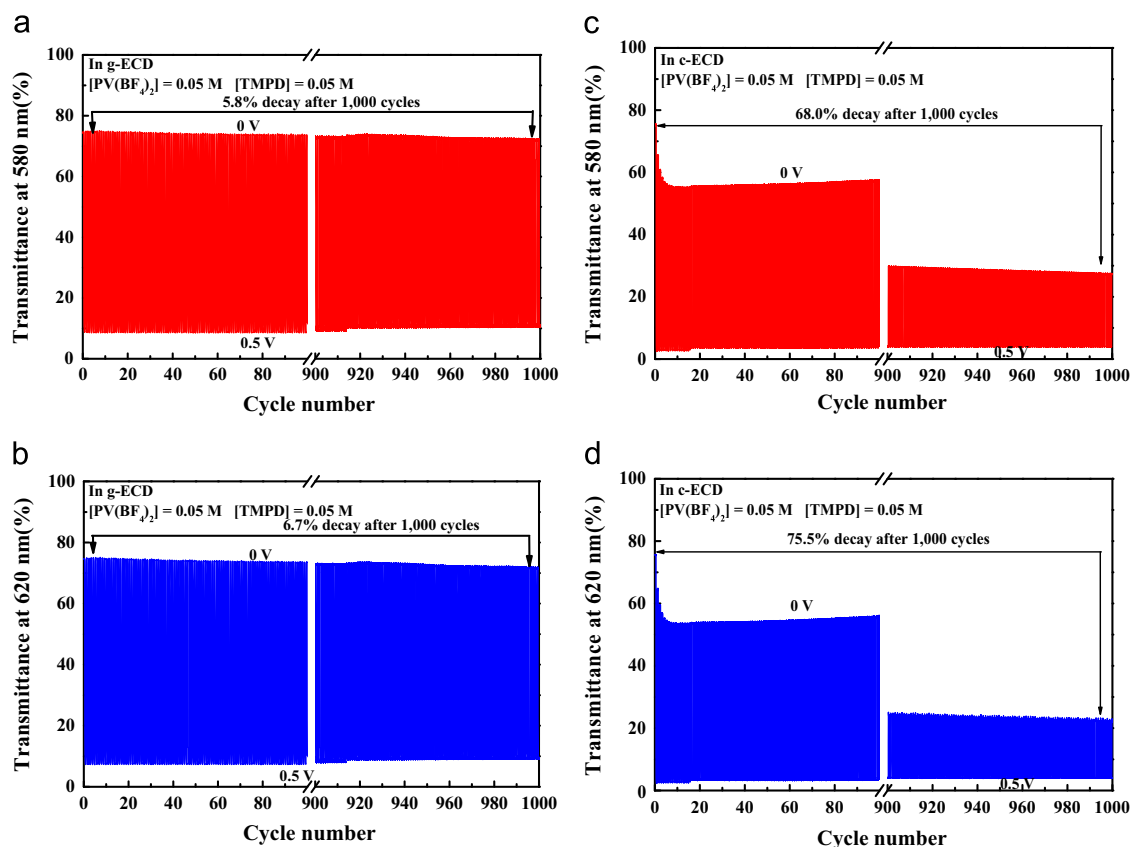


Fig. 5. Performances in write-erase ability of the g-ECD recorded at (a) 580 and (b) 620 nm. The potentials were switched between 0 and 0.5 V with the interval of 10 s for 1000 cycles. Measurements of the c-ECD recorded at (c) 580 and (d) 620 nm were also done under the same operation condition.

Table 2

Partial list of the performances of viologen-based ECDs reported in literatures.

ECD	$\lambda_{\max}^a$ [nm]	V_b/V_c^b [V]	T_b/T_c [%]	ΔT^c [%]	Cycling stability	Ref.
RGQ ^d /EV ^e +HQ ^f +BMIBF ₄ ^g /RGQ	600	0/1.5	–	27.0	–	[11]
HV ^h /TMPD	615	0/1.0	78.0/2.5	75.5	> 100 Cycles without significant decay	[14]
PACV ²⁺ /ITO	610	–2.0/2.0	90.5/51.0	39.5	> 11 Cycles without significant decay	[36]
TiO ₂ /viologen derivatives ^j /PB ^k /ATO ^l	600	2.0/–2.0	67.3/0.2	67.1	Remain 92% of its initial ΔT after 8000 cycles	[37]
PPBP ^m /PMMA–EMIB(CN) ₄ ⁿ /PB	590	1.5/–1.5	82.3/34.5	47.8	ΔT dropped from 37.8 to 32.6 after 1000 cycles	[38]
PEDOT–ViO ^o /IL gel ^p /PB	568	1.5/–1.5	–	20.0	> 11 Cycles without significant decay	[39]
IBV ^q /P(HEMA) ^r +EtMelm ^s N(CN) ₂ ^s /PB	605	0/–2.6	65.5/35	30.5	Peak current density decreased by 18.9% after 1000 cycles	[40]
WO ₃ –PBV ^t /RP ^u	580	1.0/–1.0	80.0/31.0	49.0	Remain 89% of its initial ΔT after 500 cycles	[41]
PV(BF ₄) ₂ +TMPD in polymer network	620	0/0.5	74.4/7.3	67.1	ΔT remains of 55% after 10,000 cycles	This work

^a Maximum absorbance wavelength.

^b Bleaching (V_b) and coloring potential (V_c).

^c All transmittance changes listed here have included the absorbance of two conducting substrates.

^d Reduced graphene oxide.

^e Ethyl viologen.

^f Hydroquinone.

^g 1-Methyl-3-butylimidazolium tetrafluoroborate.

^h Heptyl viologen.

ⁱ Cross-linked matrix containing viologen.

^j 1,4-Bis[[(N-Phosphono-2-ethyl)-4,4'-bipyridinium)-methyl]-benzene tetrachloride.

^k Prussian blue.

^l Antimony tin oxide.

^m Poly-(cyclotriphosphazene-4,4'-bipyridinium)chloride.

ⁿ 1-Ethyl-3-methylimidazolium tetracyanoborate.

^o N,N'-bis(3-sulfonatopropyl)-4,4'-bipyridinium doped poly(3,4-ethylenedioxythiophene).

^p Mixture of polyvinyl alcohol, 1-butyl-1-methylpyrrolidiniumtrifluoromethanesulfonate in dimethyl sulfoxide.

^q 1,1'-Bis[4-(5,6-dimethyl-1H-benzimidazole-1-yl)butyl]-4,4'-bipyridinium dibromide.

^r Poly(2-hydroxyethyl methacrylate).

^s 1-Ethyl-3-methylimidazolium dicyanamide.

^t Poly(butyl viologen).

^u Ruthenium purple.

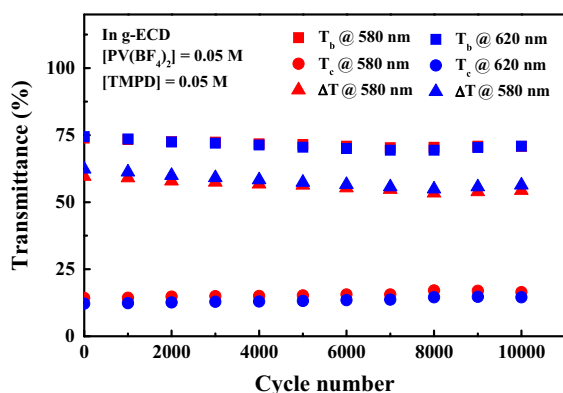


Fig. 6. Variation in transmittance of the g-ECD tested for 10,000 cycles. The g-ECD was switched between 0 and 0.5 V with the interval of 10 s.

homogeneously if uniform temperature of the whole device is maintained. The sheet resistivity of the conducting substrate should also be considered in order to scale up the ECD to the size of an electrochromic window. In this case, the I-R drop becomes more significant with increasing size of the device [35]. This infers that a higher potential bias and longer response time are required in order to achieve similar attenuations in the transmittance when the device is enlarged in size. Nevertheless, a higher potential bias would somewhat sacrifice the cycling stability [2]. Therefore, a low driving potential for the stable operation of ECDs becomes very desirable. In this work, a size of 4 cm² g-ECD requires only 0.5 V of potential bias; this suggests that a better cycling stability can be expected when compared with other relatively high voltage-driven ECDs of the same size.

A partial list of EC performances of the viologen-based ECDs reported in literatures is summarized in Table 2 [11,14,36–41]. The proposed g-ECD requires the lowest driving voltage (0.5 V) among all ECDs and still exhibits a ΔT higher than most studies. In our previous study, the viologen-based ECD showing a ΔT over 75% was proposed, but a much higher driving voltage (1.0 V) was required for that ECD [14]. Furthermore, this ECD was operated in a solution phase, in which the electrolyte leakage remains concerned. Taya et al. also proposed a viologen-based ECD configuration exhibiting a high ΔT value of 67.1%, but the high driving voltage of $-2.0/+2.0$ V is required [37]. The g-ECD proposed in this work has prevented the leakage problems while showing no significant decay in ΔT even after 3000 cycles of switching. More importantly, it requires the lowest driving voltage to operate the leakage-free, all-in-one, EC gel to date.

To investigate even longer cycling stability, a g-ECD was further subjected to 10,000 cycles of switching between 0 and 0.5 V (200,000 s in total). As suggested in Fig. 6 and Table S2, the g-ECD still achieved over 55% of ΔT at both detected wavelengths after 10,000 switching. Moreover, the response time remained below 3.5 s either for bleaching or coloring process. Similarly, the response time of c-ECD kept unchanged after 1000 cycles; however, 23.4% and 17.9% of ΔT were obtained at 580 and 620 nm, respectively, as summarized in Table S3. It proved that the g-ECD possesses excellent cycling stability. Ultimately, the huge optical contrast, low driving voltage and enhanced cycling stability enable this g-ECD to be suitable for energy-saving smart window and display applications.

4. Conclusions

A viologen-based ECD utilizing an all-in-one electrochromic gel (g-ECD) was successfully fabricated through a simple in-situ

thermal curing method. The introduced polymer network not only eliminates the leakage problem of the typical solution-type ECDs, but also significantly improves the write–erase ability compared to the conventional ECD. Besides, since the incorporation of the polymer network does not seriously retard the diffusion or migration of the electrochromic species, the obtained g-ECD can exhibit a large ΔT of 67.1% with a short response time (< 3.5 s) under an extremely low driving voltage (0.5 V). The g-ECD retains 92.8% and 92.7% of its initial ΔT after 3000 cycles of switching under 580 and 620 nm, respectively, and the g-ECD still exhibits stable EC behavior even after 10,000 cycles of switching. These desirable traits make the gelled ECD a remarkable candidate for energy-saving smart window or display applications.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.solmat.2015.04.012>.

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